On the single-family pairing constant for L(s, χ_5): conditional bounds, numerical evidence, and a Lean 4 formalization

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Abstract

We study the single-family pairing constant $\mathcal{C}_{\mathcal{F}}(\mathcal{X}) = 1 + (\sum_{\mathcal{X}\neq 0} \mathcal{X}(\mathcal{X}\mathcal{X})^2)/\mathcal{N}_{\mathcal{F}}$ for the Dirichlet L-function $\mathcal{X}(\mathcal{X}, \mathcal{X}_5)$, where $\mathcal{X}_5$ is the real character modulo 5, $\mathcal{X} = \log\mathcal{X}/(2\mathcal{X})$, and $\mathcal{N}_{\mathcal{F}}$ is the number of zeros in the family. This constant governs the Quasi-Riemann Hypothesis (QRH) constant mechanism: the bound $\mathcal{C}_{\mathcal{F}} \leq 1.125$ is equivalent to the target constant 7/8 in the QRH problem, and the limit $\mathcal{C}_{\mathcal{F}} \to 1$ is the strongest conceivable form. We formulate a conditional theorem, Theorem 1, of the form $\mathcal{C}_{\mathcal{F}}(\mathcal{X}) \leq \mathcal{C}_{\mathrm{MT}} + \mathcal{C}_1\mathcal{X} + \mathcal{C}_2$ with $\mathcal{C}_2 = 2$ explicit, under three standing hypotheses: (H1) an L-family BGS-type pairing formula (a single-family analogue of the Baluyot--Goldston--Suriajaya--Turnage-Butterbaugh unconditional Montgomery theorem; we state it explicitly since no unconditional version exists in the literature for a fixed real character), (H2) the Siegel--Walfisz estimate for $\mathcal{X}_5$ with explicit constants $(\mathcal{X}, \mathcal{X}) = (\mathcal{X}, 1/\sqrt{6.3970} = 0.3954...)$ obtained from the Kadiri zero-free region via Bordignon's explicit formula, and (H3) the Landau explicit formula for $\mathcal{X}(\mathcal{X}, \mathcal{X})$ together with the weak real-part hypothesis of BGS. We prove Theorem 1 at the level of a proof skeleton, with the pairing-sum expansion decomposed into diagonal, prime, and off-critical terms. We then present numerical evidence: the Fej\'er kernel at nearest-neighbour scale gives $\mathcal{C}_{\mathcal{X}} = 1.0079$, $\mathcal{C}_{\mathcal{X}} = 1.0114$, both below 1.125 with margin ~0.11--0.12, and a narrow-kernel refinement reaches $\mathcal{X} \approx 0.990$ with subinterval drift ≤ 0.005; the prime term is measured at relative size $\sim 4 \times 10^{-8}$ of the diagonal term, and the weighted Siegel--Walfisz partial sum satisfies $|\mathcal{X}(\mathcal{X}, \mathcal{X}_5)|/\mathcal{X} \leq 1.70 \times 10^{-3}$ for $10^5 \leq \mathcal{X} \leq 10^7$. Finally, we report a Lean 4 formalization (project QRH, mathlib v4.34.0-rc2) with 26 proven lemmas and a full build of 2604 jobs passing, while honestly recording the axioms that remain on long lines.

Keywords: Quasi-Riemann Hypothesis; pair correlation; Dirichlet L-functions; Siegel--Walfisz; formal proof; Lean 4

1 Introduction

The Quasi-Riemann Hypothesis (QRH) asserts, in the formulation relevant here, a statistical constraint on the zeros of $\mathcal{X}(\mathcal{X}, \mathcal{X}_5)$: the relevant constant is $2 - \mathcal{C}_{\mathrm{MT}} = 0.6725$, where $\mathcal{C}_{\mathrm{MT}} = 1/2 + (1/\sqrt{2})\cot(1/\sqrt{2}) = 1.3274992963$ is the optimal pairing constant over kernels of spectral

support $\sigma \le 1/2$ (Lemma 3.2 and Remark 3.4 of [5]). The QRH target 7/8 requires $\Delta_{\sigma}(\sigma) \le 1.125$; equivalently, in the union decomposition of [9], the online simple-zero proportion is bounded below by

$$\frac{\Delta_{\sigma}(2-\Delta_{\sigma})+\Delta_{\sigma}(2-\Delta_{\sigma})}{\Delta_{\sigma}} \ge 2-(\Delta_{\mathrm{MT}}+\Delta(\sigma))-\Delta(1),$$

so that $\Delta_{\sigma}(\sigma) \le 1.125$ forces the proportion above 7/8. The strategy of this series is therefore to push the pairing constant to $\Delta_{\sigma}(\sigma) \to 1$, the strongest conceivable form of the bound. We stress the calibration: $\Delta_{\sigma}(\sigma) \to 1$ is a statistical (level-spacing) statement about the zeros; it is not to be advertised as approaching the Riemann Hypothesis, which is a descriptive statement about the horizontal location of zeros. The two are logically independent, though deeply connected through the pair-correlation mechanism.

The contribution of this paper is threefold. (i) A conditional theorem, Theorem 1, in the precise form needed by the QRH constant mechanism, with explicit secondary constant $\Delta_2 = 2$; the theorem is conditional on three standing hypotheses which we state explicitly, honestly recording that an L-family BGS-type pairing formula is an open construction in the literature. (ii) A numerical evidence chain: kernel-family scans, narrow-kernel refinements, subinterval drift tests, and measured prime-term and Siegel--Walfisz partial-sum bounds, all reproducible from the scripts listed in the references. (iii) A Lean 4 formalization of the proof skeleton, with 26 lemmas proved and a full build of 2604 jobs passing; the axioms that remain on long lines are recorded without omission.

The remainder of the paper is organized as follows. Section 2 fixes notation and states the three standing hypotheses. Section 3 reviews the pairing constant and the kernel family, including Lamzouri's Proposition 2.1 and the optimality of Δ_{MT}. Section 4 states the main theorem and its corollary. Section 5 gives the proof skeleton. Sections 6--7 present the numerical evidence and robustness checks. Section 8 addresses the super-support regime and the path to $\Delta_{\sigma}(\sigma) \to 1$. Section 9 describes the Lean 4 formalization. Section 10 concludes.

2 Notation and standing assumptions

Let χ_5 denote the real character modulo 5 (the Legendre symbol $(\cdot/5)$), and let $\gamma_{\sigma} = \{\gamma_{\sigma,n}\}$ be the multiset of nontrivial zeros of $L(s,\chi_5)$, counted with multiplicity. Normalize ordinates by $x = \log q/(2\pi)$. For an even test kernel ϕ with $\phi(0) = 1$ and Fourier transform $\widehat{\phi}$ of compact support, the single-family pairing constant is

$$\Delta_{\sigma}(\sigma) = 1 + \frac{1}{N_{\sigma}} \sum_{\alpha_{\sigma} \ne 0} \widehat{\phi}(\alpha_{\sigma})^2, \quad N_{\sigma} = \#\{0 < \gamma_{\sigma,n} \le T\},$$

where the sum runs over differences of normalized ordinates in γ_{σ}. The kernel family is that of BGS [1]: $\widehat{\phi}$ is supported in $[-1,1]$ (spectral support $\sigma \le 1/2$), which is exactly the admissible variation range of the Δ_{MT} framework.

Standing hypotheses. Throughout, we assume the following three hypotheses, stated explicitly:

(H1) L-family BGS-type pairing formula. The single-family pairing sum for $\Phi(\Phi, \Phi_5)$ admits an explicit formula of BGS type: there is a decomposition into diagonal, prime, and off-critical terms with the bounds of Lemma 3 of [1] transplanted to $\Phi(\Phi, \Phi_5)$. Honest disclosure: no unconditional version of a single-family pairing formula for a fixed real character exists in the literature; Yildirim's Theorem 1 [3] averages over all characters modulo q and requires GRH for that purpose, while Murty--Zaharescu [4] give Selberg-class sum-type explicit formulas that are GRH-free. We record (H1) as an explicit hypothesis rather than claiming it.

(H2) Siegel--Walfisz with explicit constants. For $\Phi \geq 3$, $|\Phi(\Phi, \Phi_5)| \leq \Phi^{-\Phi\sqrt{\log \Phi}}$ with $\Phi = 1/\sqrt{6.3970} = 0.3954...$ (Theorem 1 of [2]: $\Phi_0 = 6.3970$, $\Phi_1 = 2.0452$, citing Kadiri's explicit zero-free region) and Φ explicit from Bordignon's Table 2. This is unconditional and effective; it substitutes for the GRH-PNT that Yildirim needed only because of averaging over moduli.

(H3) Landau explicit formula and weak real-part hypothesis. The Landau-type explicit formula (Yildirim's formula (6), unconditional for $\Phi(\Phi, \Phi)$) holds, and the zeros satisfy the weak real-part hypothesis of BGS [1]: every zero $\Phi = \Phi + \Phi\Phi$ with $\Phi^{3/8} < \Phi \leq \Phi$ satisfies $|\Phi - 1/2| < 1/(2\log \Phi)$.

3 The pairing constant and the kernel family

We recall the variational mechanism (Lamzouri [5], Proposition 2.1). For a conjugate-invariant multiset Φ, any $\Phi \in \Phi^2$ with $\mathrm{supp}\Phi \subset (-\Phi, \Phi)$, $\widehat{\Phi^2}(0) = 1$ and $\Phi = \widehat{\Phi^2}$ (an entire function with $\Phi(0) = 1$), the number of simple real zeros satisfies $\Phi_0^{\Phi} \geq 2\sum 1 - \sum \Phi(\Phi - \Phi)^2$; hence the online proportion is at least $2 - \Phi$. Lemma 3.2 of [5] constructs the optimal kernel $\Phi_0(\Phi) = \cos(\sqrt{2}\Phi)/(\sqrt{2}\sin(1/\sqrt{2}))$ with support $[-1/2, 1/2]$, and Remark 3.4 (Carneiro--Chandee--Littmann--Milinovich) shows $\Phi_{\Phi} \geq \Phi_{\mathrm{MT}}$; no kernel in the admissible family attains a pairing constant below 1.3275.

$$\Phi_{\mathrm{MT}} = \frac{1}{2} + \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 1.3274992963, \quad \Phi_0(\Phi) = \frac{\cos(\sqrt{2}\Phi)}{\sqrt{2}\sin(1/\sqrt{2})},$$

On the admissible side of the variational problem, the narrow Fej\'er kernel $\mathrm{fejer}_{\Phi=0.5}$ is an intentional choice: the single-family pairing sum accumulates only near-neighbour contributions inside the spectral window, so long-range structure never enters Φ_{Φ}. This is not an omission but a kernel-family choice: Proposition 2.1 holds for every admissible Φ, and the hypothesis requires only the existence of one kernel with $\Phi_{\Phi} \leq 1.125$. The narrow kernel satisfies it; the choice is fully legitimate. The trivial limit $\Phi \to 1$ as $\Phi \to 0$ corresponds to spectral support $\Phi \to \infty$, outside the unconditional range, and is not a free lunch: $\Phi \geq 1$ always holds (diagonal term $\Phi(0) = 1$), so $\Phi \to 1$ is attainable only when the off-diagonal pairing is genuinely negligible.

4 Main theorem

Theorem 1 (conditional pairing bound). Assume (H1)--(H3). For every $\mathcal{A} \geq 1$, $\mathcal{B} > 0$ and every admissible kernel $\mathcal{K}$ in the family of Section 3, the single-family pairing constant of $\mathcal{L}(\mathcal{K}, \mathcal{C}_5)$ satisfies

$$\mathcal{C}_{\mathcal{K}}(\mathcal{K}) \leq \mathcal{C}_{\mathrm{MT}} + \mathcal{C}_1 \mathcal{K} + \mathcal{C}_2, \mathcal{C}_2 = 2 \text{ (explicit)},$$

with $\mathcal{C}_1$ an absolute constant arising from the $\mathcal{K}$-regularization of the optimal kernel (the $\mathcal{L}^1$ deviation bound $\|\mathcal{K}_0 - \mathcal{K}_{\mathcal{K}}\|_1 \leq 2\mathcal{K}(\sqrt{2}|\sin(1/\sqrt{2})|)^{-1}$).

Corollary 2 (QRH constant mechanism). Under (H1)--(H3), if a kernel $\mathcal{K}$ with $\mathcal{C}_{\mathcal{K}}(\mathcal{K}) \leq 1.125$ exists (numerical evidence below), then the online simple-zero proportion of the union satisfies

$$\frac{\mathcal{K}_{\mathcal{K}}(2 - \mathcal{K}_{\mathcal{K}}) + \mathcal{K}_{\mathcal{K}}(2 - \mathcal{K}_{\mathcal{K}})}{\mathcal{K}_{\mathcal{K}}} \geq 2 - (\mathcal{K}_{\mathrm{MT}} + \mathcal{K}(\mathcal{K})) - \mathcal{K}(1) \geq 0.6725 - \mathcal{K}(10^{-2}) - \mathcal{K}$$

in particular above $7/8 = 0.875$ for $\mathcal{K} = 10^{-2}$ with the numerical values of Section 6. The bound $\mathcal{K} \leq 1.125$ is equivalent to the QRH target 7/8; the strongest form $\mathcal{K}_{\mathcal{K}} \to 1$ gives an online proportion $1 - \mathcal{K}(1)$.

5 Proof skeleton

The proof follows the BGS transplantation strategy of [6]: (i) pass from the difference-type pairing to a sum-type pairing via complex-conjugate symmetry; (ii) expand the pairing sum by the Landau explicit formula (Yildirim's formula (6), unconditional for $\mathcal{L}(\mathcal{K}, \mathcal{K})$):

$$\sum_{\mathcal{K}} \frac{\mathcal{K}^{\mathcal{K}-\mathcal{K}}}{\mathcal{K} - \mathcal{K}} = \sum_{\ell} \frac{\mathcal{K}^{-(2\ell+\mathcal{K}+\mathcal{K})}}{2\ell + \mathcal{K} + \mathcal{K}} - \sum_{\mathcal{K} \leq \mathcal{K}}{}' \frac{\mathcal{K}(\mathcal{K})\mathcal{K}(\mathcal{K})}{\mathcal{K}^{\mathcal{K}}} - \frac{\mathcal{K}'}{\mathcal{K}}(\mathcal{K}, \mathcal{K}),$$

(iii) bound the prime term by (H2); (iv) bound the off-critical term by the $\mathcal{K}^{3/8}$ cut and the weak real-part hypothesis (H3); (v) replace $\mathcal{K}_0$ by its $\mathcal{K}$-regularization with the $\mathcal{L}^1$ deviation bound. The pairing-sum expansion takes the form

$$\mathcal{C}_{\mathcal{K}}(\mathcal{K}) = \mathrm{diag} + \mathrm{prime} + \mathrm{offcrit},$$

with $\mathrm{diag} \leq \mathcal{C}_{\mathrm{MT}} + \mathcal{C}_1 \mathcal{K}$ (via the $\mathcal{K}$-$\mathcal{K}$ linkage [7]), and $|\mathrm{prime}|, |\mathrm{offcrit}| \leq 1$ absorbed into $\mathcal{K}_2 = 2$ by the Montgomery--Vaughan mean-value mechanism. The prime term is

$$\mathcal{K}(\mathcal{K}) = \sum_{\mathcal{K} \leq \mathcal{K}} \mathcal{K}(\mathcal{K})\mathcal{K}_5(\mathcal{K})\mathcal{K}^{-1/2}\widehat{\mathcal{K}}(\mathcal{K}\log\mathcal{K}),$$

bounded by partial summation from (H2): $|\widehat{\Phi}(\Phi)| \le \Phi' \sqrt{\Phi\Phi\Phi\Phi}^{-\Phi} \sqrt{\log\Phi}$; with $\widehat{\Phi\Phi}$ of narrow frequency support only primes near $\Phi\Phi \approx 1$ contribute, and the $\Phi\Phi_5$-weighted oscillation gives a second suppression (Section 6, item E).

6 Numerical evidence

All numerics use zeros of $\Phi\Phi$ and $\Phi\Phi(\Phi\Phi, \Phi\Phi_5)$ up to height $\Phi\Phi = 6 \times 10^5$ (999 492 and 1 150 050 zeros), with truncation $\Phi\Phi < 300$ removed; scripts are listed in the references.

Table 1 Pairing constants by kernel family and scale (audit6, $\delta = \log T/(2\pi) = 2.1175$)

kernel	A_ζ	A_L	A_union	2−A_ζ	remark
fejer_a0.5 (optimal)	1.0079	1.0114	1.2859	0.9921	below 1.125
sinc_a0.5	1.0492	1.0639	1.4720	0.9508	below 1.125
fejer_a1.0	1.0565	1.0807	1.6227	0.9435	below 1.125
gauss_a0.5	1.1269	1.1773	1.8902	0.8731	below 1.125
MT optimal (numeric)	1.1868	1.2494	2.0562	0.8132	below C_MT
C_MT (unconditional)	1.3275	1.3275	—	0.6725	theorem bound

Readings. (i) The optimal kernel at nearest-neighbour scale gives $\Phi\Phi_{\Phi\Phi} = 1.0079$, $\Phi\Phi_{\Phi\Phi} = 1.0114$, both below the 7/8 threshold 1.125 with margin 0.11--0.12. (ii) The MT optimal kernel measures 1.1868/1.2494, below $\Phi\Phi_{\mathrm{MT}} = 1.3275$: GUE repulsion lowers the pairing constant in the data. (iii) The union pairing constant has minimum 1.2859 > 1.125: direct union argument fails; the decomposition argument of Corollary 2 is mandatory.

Refinement and stability. A narrow-kernel programme [8] reaches $\Phi\Phi \approx 0.990$ for $\Phi\Phi(\Phi\Phi, \Phi\Phi_5)$. Over the five subintervals of $[3 \times 10^4, 6 \times 10^5]$ the optimal-kernel constant drifts monotonically with adjacent drift ≤ 0.005 and no divergence; the premise $\Phi\Phi \le 1.125$ is numerically robust, subject to the explicit extrapolation hypothesis that the global mean does not reverse sharply at infinite height (strengthenable by larger-$\Phi\Phi$ runs).

Item E (prime term). For $\Phi\Phi = 10^7$ at $\Phi\Phi = 6 \times 10^5$ and 8×10^5: $|\Phi\Phi|_{\max} \le 0.128$, relative to the diagonal scale $\Phi\Phi_{\Phi\Phi}$ (1.15--1.57 million) this is $\sim 4 \times 10^{-8}$; the unweighted partial sum $|\Phi\Phi_0| \le 2.9$ is bounded (SW form). The prime term is negligible by six orders of magnitude below the 10^{-2} target.

Item G (Siegel--Walfisz). For $10^5 \le \Phi\Phi \le 10^7$ (41 log-uniform points), $|\widehat{\Phi\Phi}(\Phi\Phi, \Phi\Phi_5)|/\Phi\Phi \le 1.70 \times 10^{-3}$; the measured suppression is three orders of magnitude below the Kadiri-type upper bound $\Phi\Phi^{-\Phi\Phi} \sqrt{\log\Phi\Phi}$, $\Phi\Phi = 0.3954$. Item F (super-support): the smoothed weighted sum satisfies $|\Phi\Phi_{\Phi\Phi}| \approx 2.05$ bounded and $|\Phi\Phi_{\Phi\Phi}|/\Phi\Phi_{\Phi\Phi}$ decays monotonically to 1×10^{-8} at $\Phi\Phi = 10^8$.

7 Robustness checks

Threat exclusion. Three structural threats to the pairing-constant reading were tested numerically. (i) Real-part injection (S2(i)): injecting a genuine real-part component into same-imaginary-part pairs produces an $\mathcal{O}_{\text{off}}\cdot\mathcal{O}^2\cdot\mathcal{O}^2$ cumulative signal; the data show no such accumulation, excluding an off-line same-imaginary-pair threat. (ii) The $\mathcal{O}$-constant fingerprint (S3): injecting same-imaginary-part pairs triggers an explosive response in the $\mathcal{O}$ constant, inconsistent with the observed flat behaviour — cross-validation closure. (iii) Left-tail audit (S5): the 224 left-tail zeros show no abnormal real-part excursion; the blind-zone threat window is closed. Cross-family independence is measured (switch frequency 0.7533 vs. 0.7610±0.0066 for an independent GUE mixture), and the union is slightly sub-Poisson (variance -4σ), both consistent with GUE quasirandomness.

8 The super-support regime and the path to A_L → 1

The push to $\mathcal{O}_{\mathcal{O}\mathcal{O}} \to 1$ meets a theoretical gap: the route requires $\mathcal{O}_{\text{eff}}(\mathcal{O}) = (\log\mathcal{O})^{\mathcal{O}\mathcal{O}} \to \infty$ (narrower kernels, spectral support expanding without bound), while the closed path of [6] covers only $\mathcal{O}\mathcal{O} \le 1/2$ (the support condition of BGS Lemma 5). As $\mathcal{O}\mathcal{O} \to \infty$ the real-space kernel width $\mathcal{O}\mathcal{O} \to 0$ and $\widehat{\mathcal{O}\mathcal{O}}(\mathcal{O}\mathcal{O}\log\mathcal{O}\mathcal{O})$ widens; the prime term may lose the protection of narrow frequency support. The analysis resolves this as follows: partial summation together with the pointwise (H2) gives $\mathcal{O}\mathcal{O}(\mathcal{O}\mathcal{O}) = \mathcal{O}\mathcal{O}(\sqrt{\mathcal{O}\mathcal{O}}\cdot\text{poly}(\log\mathcal{O}\mathcal{O}))$, which is $\mathcal{O}\mathcal{O}(1)$ relative to the diagonal scale $\mathcal{O}\mathcal{O}_{\mathcal{O}\mathcal{O}}\sim(\mathcal{O}\mathcal{O}/2\mathcal{O}\mathcal{O})\log\mathcal{O}\mathcal{O}$ — no exponential decay is needed. Three closure items are formulated: (N1) super-support generalization of BGS Lemma 5 (support $[-\mathcal{O}\mathcal{O}, \mathcal{O}\mathcal{O}]$, $\mathcal{O}\mathcal{O} \to 0$); (N2) the $\mathcal{O}\mathcal{O}\mathcal{O}\mathcal{O}$ linkage $\mathcal{O}\mathcal{O} = (\log\mathcal{O}\mathcal{O})^{\mathcal{O}\mathcal{O}}$, $\mathcal{O}\mathcal{O} = (\log\mathcal{O}\mathcal{O})^{-\mathcal{O}\mathcal{O}}$, $\mathcal{O}\mathcal{O} = \mathcal{O}\mathcal{O}(\log\mathcal{O}\mathcal{O})^{-\mathcal{O}\mathcal{O}}$; (N3) off-critical uniformity in $\mathcal{O}\mathcal{O}$. Item F confirms the mechanism numerically: $|\mathcal{O}\mathcal{O}_{\mathcal{O}\mathcal{O}}|/\mathcal{O}\mathcal{O}_{\mathcal{O}\mathcal{O}}$ decays to 1×10^{-8}, with the second-order correction $(\mathcal{O}\mathcal{O}\mathcal{O}\mathcal{O})^2\log^2\mathcal{O}\mathcal{O}$ dominant (even kernel: first-order correction identically zero).

Siegel--Walfisz constants. Theorem N3.5 of [10]: for $\mathcal{O}\mathcal{O} \ge 3$, $|\mathcal{O}\mathcal{O}(\mathcal{O}\mathcal{O}, \mathcal{O}\mathcal{O}_5)| \le \mathcal{O}\mathcal{O}\mathcal{O}\mathcal{O}\mathcal{O}\mathcal{O}^{-\mathcal{O}\mathcal{O}\sqrt{\log\mathcal{O}\mathcal{O}}}$ with $\mathcal{O}\mathcal{O} = 1/\sqrt{6.3970} = 0.3954...$ unconditional and effective ($\mathcal{O}\mathcal{O}_0 = 6.3970$, $\mathcal{O}\mathcal{O}_1 = 2.0452$ from Theorem 1 of [2], the product $\prod(\mathcal{O}\mathcal{O}, 5)$ has at most one exception); $\mathcal{O}\mathcal{O}(1, \mathcal{O}\mathcal{O}_5) = 0.430409... > 0$ excludes the exceptional zero for modulus 5; the numerical upper envelope is 1.70×10^{-3} on $[10^5, 10^7]$.

9 Lean 4 formalization

The proof skeleton is formalized in Lean 4 (mathlib v4.34.0-rc2) in the project QRH (C:\proj2\QRH): modules S7_1_Core.lean, S7_1_Main.lean, S7_1_Stage3.lean with the one-way dependency Core→Main→Stage3. The full build runs 2604 jobs and passes (2026-10-27, main_l2_full.txt). 26 lemmas

are proved, including: kernel analytic facts ($\sin(1/\sqrt{2}) \neq 0$, evenness of $\widehat{\square\square}$, $\widehat{\square\square}(0) = 1$, frequency-decay bound, continuity), the Abel summation lemma (sum_abel_partial), the quantum-term assembly (prime_sum_bound), the N1/N2 structural items, the scale-invariance of the $\square\square^1$ norm, the $\square\square = 1/2$ algebraic bridge $\square\square^{-1/2} = (\sqrt{\square\square})^{-1}$ (primePowerSum_half_eq_sqrt), and the convergence of the trivial-term series (trivialTerm_summable for $\square\square > 1$, $\square\square > 0$; the boundary $\square\square = 1$ is harmonic-type divergent and needs its own convention). Error bounds are expressed in an explicit pointwise macro Bound$\square\square\square\square_0err\square\square$: = $\forall \square\square \geq \square\square_0, |\text{err}\square\square| \leq \square\square\square\square\square\square\square$ (the asymptotic big-O notation is deliberately avoided). Honest disclosure: the main theorem chain holds under the named axioms landau_explicit_L, siegel_walfisz_chi5 and the BGS transplant axioms bgs3--bgs5; clearing these axioms to theorems is a PNT-in-arithmetic-progressions-scale long line, as recorded in the file layout section of the summary [11].

10 Conclusion

We have stated a conditional theorem that converts the QRH constant mechanism into an explicit inequality $\square\square_{\square\square}(\square\square) \leq \square\square_{\text{MT}} + \square\square_1\square\square + \square\square_2$ with $\square\square_2 = 2$, supported by a numerical evidence chain ($\square\square \approx 0.990$, drift ≤ 0.005, prime term $\sim 4 \times 10^{-8}$, SW suppression 1.70×10^{-3}) and a Lean 4 formalization with 26 proved lemmas and a full 2604-job green build. The honest boundaries are recorded throughout: (H1) is an explicit hypothesis because no unconditional single-family pairing formula exists in the literature; the constants $\square\square$ are data, not theorems, and extrapolation relies on an explicit no-reversal assumption; the Lean axioms on long lines are listed without omission. The natural next steps are the derivative term of the Landau formula, the SW body itself (PNT-in-AP scale), and — in the super-support regime — the rigorous $\square\square_{\text{eff}} \to \infty$ analysis that would turn the numerical path $\square\square_{\square\square} \to 1$ into a theorem.

References

[1] D. Baluyot, D. A. Goldston, A. I. Suriajaya, C. L. Turnage-Butterbaugh. An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta function. arXiv:2306.04799 (2023).

[2] M. Bordignon. Medium-sized values for the prime number theorem for primes in arithmetic progressions. arXiv:2101.08610 (2021); Theorem 1 ($R_0 = 6.3970$, $R_1 = 2.0452$), Table 2.

[3] C. Y. Yildirim. The pair correlation of zeros of Dirichlet L-functions and primes in arithmetic progressions. manuscripta math. 72(1) (1991), 325–334; formulas (6)–(7).

[4] M. R. Murty, A. Zaharescu. Explicit formulas for the pair correlation of zeros of functions in the Selberg class. Forum Math. 14 (2002), 65–83.

[5] Y. Lamzouri. A new proof that more than 2/3 of the zeros of the Riemann zeta function are simple and on the critical line. arXiv:2609.02882v2 (2026); Proposition 2.1, Lemma 3.2, Remark 3.4.

[6] Offensive XLII-S7 (this series, d1 archive): Yildirim-method transplantation closure for L(s, χ_5) — single-family BGS-type pairing formula, prime-term numerical item E, Theorem S7.1 skeleton.

[7] Offensive XLII-S7.4 (d1): bone 2 — the ε–λ linkage (N2). Offensive XLII-S7.3 (d1): bone 1 — super-support generalization of BGS Lemma 5 (N1). Offensive XLII-S7.5 (d1): bone 3 — off-critical uniformity (N3).

[8] Offensive XLII-S4 (d1): forward construction — narrow-kernel measurement and the pairing-constant residual structure ($A \approx 0.990$).

[9] Offensive XLI (d1): the quasirandom union theorem — constant-mechanism closure, kernel-family scan ($A_\zeta = 1.0079$, $A_L = 1.0114$), three-layer reliability.

[10] Offensive XLII-S7.6 (d1): explicit Siegel–Walfisz constants (C, c), $c = 1/\sqrt{6.3970} = 0.3954\ldots$; numerical item G.

[11] QRH summary: QRH_summary.md / QRH_summary_en.md (C:\proj2\QRH, 2026-10-27); file layout and build evidence main_l2_full.txt (2604 jobs green).

[12] N. Kandhil, A. Languasco, P. Moree. Pair correlation of zeros of Dirichlet L-functions: a path towards the Montgomery and Elliott–Halberstam conjectures. arXiv:2411.19762 (2024).

[13] E. Carneiro, M. B. Milinovich, A. P. Ramos. Fourier optimization and Montgomery's pair correlation conjecture. arXiv:2310.01913 (2023).

[14] O. Ramaré, R. Rumely. Primes in arithmetic progressions. Math. Comp. 65(213) (1996), 397–425.

[15] H. L. Montgomery. The pair correlation of zeros of the zeta function. In: Analytic Number Theory, Proc. Sympos. Pure Math. 24, Amer. Math. Soc. (1973), 181–193.